

Naive Set Theory

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1 Sets, Elements and Axiom of Extension

1.1 What is a set?

Informally speaking, students in a classroom, pencils in a case and books in a library are all simple examples of sets. However, in the set theory we do not define the word *set*, because every theory need some elementary concepts.

Let's suppose that we had defined the concept of sets. Then, we would need some other concepts, and for defining them, we would need some other concepts and so on...

Since this process cannot goes to infinity, we need some elementary concepts that we accept them without any definition. In set theory, there are two undefined concepts: One is *set* which is an object and the other one is *belonging* which is a relation between two sets.

If x belongs to A , we say x is an element of A and write

$$x \in A.$$

Similarly, if x does not belong to A , we say x is not an element of A and write

$$x \notin A.$$

In our notes, just for convenience, we will use small case letters for elements of sets which will be denoted by upper and more complex letters. For example:

$$x \in A$$

and

$$A \in \mathcal{F}.$$

1.2 Subsets

If for all x , if $x \in A$ then $x \in B$; then we say that A is a *subset* of B and denote with

$$A \subseteq B$$

or

$$B \supseteq A.$$

Similarly, if A is not a subset of B , we say A is not a subset of B and write

$$A \not\subseteq B.$$

Now, we can prove simple theorems including subsets.

Theorem 1.1. *Every set is subset of itself.*

Proof. Since every element in A is an element of A , A is an subset of A . $\square$

Theorem 1.2. *Suppose A is a subset of B and B is a subset of C . Then, A is a subset of C .*

Proof. Suppose x is an arbitrary element in A . Since $A \subseteq B$, $x \in B$. And since $B \subseteq C$, $x \in C$. Therefore, $A \subseteq C$. $\square$

1.3 Set Equality

We have defined the concept of subsets. We also need to determine when two sets are *equal*. Since equality is a meta-concept, we need an axiom to define equality in sets.

(Axiom of Extension). *Two sets are equal if and only if they have the same elements.*

Or formally,

$$\forall A \forall B [A = B \iff \forall x (x \in A \iff x \in B)].$$

Or simply we can say, *sets are determined by their elements.*

We say that A is a *proper subset* of B , if A is a subset of B , but A and B are not equal, and denote with

$$A \subset B.$$

Theorem 1.3. *Suppose A is a subset of B and B is a subset of A . Then, A and B are equal.*

Proof. Since $A \subseteq B$, every element in A must be in B . And since $B \subseteq A$, every element in B must be in A . Thus, they have same elements. Therefore, $A = B$. $\square$

2 Creating Sets

2.1 Axiom of Existence

We talked about sets and their equality, but we did not officially say there exists a set. However, if we will work with sets, it is crucial.

There exists a set.

It is a good thing to know that there exists a set, but we do not know anything else. Are there multiple sets or there is a unique one, what are its elements, and can we sure if there is a set, then all subsets of it are also exists?

2.2 Axiom of Specification

If we want to talk about more sets or create new ones using the existing sets, we need some other axioms.

(Axiom of Specification). *Suppose A is a set and $S(x)$ is a sentence. Then, there exists a set whose elements are in A for which makes $S(x)$ true.*

Informally speaking, this axiom allows us to create subsets for any existing set. Now, we can prove some elementary facts about set theory.

Theorem 2.1 (Empty Set). *There exists a set with no elements.*

Proof. Since we know there exists a set, let A be a set. Let $S(x)$ be the sentence “ $x \neq x$ ”. If we apply the *Axiom of Specification* for A and $S(x)$, the new set must have no elements since no x satisfies $x \neq x$. $\square$

Theorem 2.2. *There exists a **unique** set with no elements.*

Proof. Since we know there exists a set with no elements, let $\emptyset_1$ and $\emptyset_2$ be the sets with no elements. According to the *Axiom of Extension*, if these sets are not equal then there must be some element in one of them which the other one does not have. But we know that these two sets have no elements. Therefore, they must be equal. $\square$

From now on we will call this set as *empty set* and denote with

$$\emptyset.$$

If A is a set and $S(x)$ is a sentence, we know that we can form a set using the *Axiom of Specification*. Let B be that set. From now on, we will use the notation

$$B = \{x \in A : S(x)\}$$

to say B is a set whose elements are in A for which make $S(x)$ true.

Theorem 2.3. *Empty set is a subset of every set.*

Proof. Let A be some set. Suppose that $\emptyset$ is not a subset of A . Then there exist some $x \in \emptyset$ for which $x \notin A$. Since $\emptyset$ has no elements, this is impossible. Thus, $\emptyset \subseteq A$. Since A was an arbitrary set, we can conclude that $\emptyset$ is a subset of every set. $\square$

Theorem 2.4. *There are no such set which have all sets as an element.*

Proof. Let A be a set. Using the *Axiom of Specification* we can construct the set $B = \{x \in A : x \notin x\}$. Since $B \in B$ lead us to $B \notin B$, which is a contradiction, $B \notin B$. Then, either $B \notin A$ or $B \in B$ (negation of $x \notin x$ for B). Since the latter is impossible, $B \notin A$. A was an arbitrary set. Therefore, for all sets, there exists a set which does not belong to the set. Hence, no set can contain all sets. $\square$

This so-called set (a set that contains everything) was called as *universal set* in early practices of set theory. This type of sets could cause a paradox named *Russell's Paradox*. For more information: https://en.wikipedia.org/wiki/Russell's_paradox

2.3 Axiom of Pairing

What we know so far is there exists some set and there is an empty one. If we want to prove more theorems about sets, we need more axioms. For example, if we know that $a_0, a_1, \dots, a_n$ are sets, we want to construct the set $A = \{a_0, a_1, \dots, a_n\}$. We need some axioms to achieve this. For start, we will order an axiom to make sure that for every set A and B , there exists a set which contain both A and B .

(Axiom of Pairing). *For all sets A and B , there exists a set which contain both A and B .*

Note that this axiom is not saying this set only contains A and B , it may have some other unwanted elements. If we apply *Axiom of Specification* with the sentence $S(x) : "x = A \text{ or } x = B"$, we get wanted result. *Axiom of Extension* guaranties its uniqueness.

If x and y are different sets we will write $\{x, y\}$ to imply this is a set which contains only x and y and say this set is *unordered pair* or just *pair* of x and y .

Since the axiom does not imply that, we can choose our two sets to be equal. In that case, the set which was created by *Axiom of Pairing* have a unique element. Then, we write $\{x\}$ to imply pair of x and x and say this set is a *singleton* of x .

2.4 Unions and Intersections

Suppose that A and B are sets. Naturally, we want to unite the elements of A and B into a single set. Since we cannot construct a set based on the axioms we have, we need some other axioms.

(Axiom of Unions). *Suppose that $\mathcal{C}$ is a collection of sets.¹ Then, there exists a set which contains all x for which, x is in at least one element of $\mathcal{C}$.*

Since *Axiom of Unions* says “which contains” instead of “only contains”, sets that are created with *Axiom of Unions* may have unwanted elements (there are no sets in $\mathcal{C}$ which contains them). If we use the *Axiom of Specification*, we can specify the elements that are in at least one set in $\mathcal{C}$.

Let $\mathcal{C}$ be a collection of sets. According to the *Axiom of Unions* there exists a set $\mathcal{U}$ which have all x in at least one set in $\mathcal{C}$. Then, $\cup\mathcal{C} = \{x \in \mathcal{U} : \text{for some } A \text{ in } \mathcal{C}, x \in A\}$ will be the set whose elements are in some set in $\mathcal{C}$. Similarly, we can say that if $x \in A$ for some $A \in \mathcal{C}$, then $x \in \cup\mathcal{C}$. This set $\cup\mathcal{C}$ is called the union of the collection $\mathcal{C}$ of sets, and the *Axiom of Extension* guaranties its uniqueness.

Instead of $\cup\mathcal{C}$, some mathematicians use

$$\cup\{X : X \in \mathcal{C}\}$$

or

$$\cup_{X \in \mathcal{C}} X.$$

Theorem 2.5. $\cup\emptyset = \emptyset$.

Proof. Suppose that $x \in \cup\emptyset$. Then, for some $A \in \emptyset$, $x \in A$. But, it is impossible to $A \in \emptyset$. Thus, $\cup\emptyset = \emptyset$. $\square$

Theorem 2.6. $\cup\{A\} = A$.

Proof. ($\subseteq$) Let $x \in \cup\{A\}$. Then, for some $X \in \{A\}$, $x \in X$. Since $\{A\}$ have a unique element namely A , $x \in A$. Thus, $\cup\{A\} \subseteq A$.

($\supseteq$) Let $x \in A$. Since $A \in \{A\}$, $x \in \cup\{A\}$. Thus, $A \subseteq \cup\{A\}$.

Since $\cup\{A\} \subseteq A$ and $A \subseteq \cup\{A\}$, $\cup\{A\} = A$. $\square$

¹Actually, everything in the set theory are sets, but if we want to emphasise that we will use some properties of sets in some set, we say *collection of sets* instead “set of sets”. Thus, this is purely for readability, not because collections and sets are different type of objects.

Definition. $A \cup B := \cup\{A, B\}$.